MADHU PRIYA P

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Profile

Final-year Electronics and Communication Engineering student with a strong interest in core electronics, embedded systems, and VLSI design. Skilled in problem-solving, analytical thinking, and applying engineering concepts to practical projects. Familiar with programming, circuit design, and technical tools through academic work and hands-on learning. Quick learner with a positive attitude, eager to contribute and grow in a professional environment. Seeking an opportunity to begin a career where I can build technical expertise and deliver value.

Skills & Abilities

- Embedded Systems
- VLSI Basics
- Digital Electronics
- C Programming
- MATLAB / Simulink
- Verilog / HDL Basics
- IoT Fundamentals
- Python
- Teamwork
- Problem Solving

Experience

Practical Training Intern – VLSI Design | Pumo Technovation | 2024 (1 Week)

- Attended one-week practical training focused on VLSI design fundamentals, semiconductor concepts, and digital IC design flow.
- Learned the basics of front-end and back-end VLSI processes, including design stages from specification to layout.
- Gained exposure to CMOS technology, logic gates, combinational and sequential circuit design concepts.
- Practiced Verilog HDL basics for designing simple digital circuits and understanding hardware description methodology.
- Worked with simulation tools to analyze circuit behavior, verify logic functionality, and observe timing results.
- Understood the importance of low power, area optimization, and performance considerations in chip design.
- Strengthened knowledge of digital electronics and expand interest in semiconductor and chip design technologies.

Embedded Systems Intern | Robomiracle | 2025

- Worked on microcontroller-based embedded system projects involving sensor interfacing, actuator control, and basic automation applications.
- Experimented and tested Embedded C programs for controlling hardware peripherals such as GPIO, timers, and input/output modules.
- Interfaced sensors and actuators to build real-time responsive systems and improve system functionality.
- Performed debugging, troubleshooting, and hardware testing to identify faults and ensure smooth operation of embedded applications.
- Gained hands-on experience in circuit connections, module integration, and practical implementation .

ACADEMIC PROJECTS

Smart Parking System

- planned and enhanced an automated smart parking system to manage vehicle entry, exit, and parking slot monitoring efficiently.
- Implemented RFID-based authentication to allow access only for authorized users and improve security.
- Integrated IR sensors to detect vehicle presence and identify available parking spaces in real time.
- Programmed gate automation using a servo motor controlled through Embedded C logic on ESP32.
- Displayed live parking status, slot availability, and access information on an LCD module.
- Improved parking management by reducing manual intervention and enabling faster vehicle movement.

Smart Shopping Cart with Budget Control System

- Created and improved an intelligent shopping cart system to simplify supermarket purchases and reduce checkout waiting time.
- Implemented real-time billing using an ESP32 microcontroller integrated with a barcode scanner to scan product barcodes instantly.
- Programmed the system to calculate and update the total bill dynamically as items were added to the cart.
- Added a user-defined budget control feature that continuously compared the total amount with the preset spending limit.
- Enhanced an automatic purchase lock mechanism that restricted additional purchases when the budget limit was exceeded.
- Integrated LCD display and buzzer alerts to notify users visually and audibly about spending status and budget limits.

Certifications

- **Generative AI Certification:** Learned basics of AI tools and how to use them to generate content and automate tasks**.**
- **Python Certification:** Learned the fundamentals of Python programming, including data types, loops, functions, and basic problem-solving for automation and software development.
- **Embedded Systems Certification (NIELIT):** Learned the fundamentals of embedded systems, microcontrollers, sensor interfacing, Embedded C programming, and real-time hardware applications.
- **VLSI Beginners Certification(NIELIT) :** Learned the basics of VLSI design, digital logic circuits, semiconductor concepts, CMOS technology, and introductory Verilog HDL for chip design.

Education

BACHELOR OF ENGINEERING IN ELECTRONICS AND COMMUNICATION ENGINEERING | 2023-2027 | NGP INSTITUTE OF TECHNOLOGY, COIMBATORE, TAMILNADU – with 76%

HSC |2023| SRI VIDHYA BARATHI MATRIC HR.SEC.SCHOOL, TIRUCHENGODE, TAMILNADU - 81%

SSLC|2021| SRI VIDHYA BARATHI MATRIC HR.SEC.SCHOOL, TIRUCHENGODE, TAMILNADU-PASS

Activities and Interests

Books – Fiction, Sports – Badminton, Current affairs.